DDD-W-80D
October 12, 1976
SUPERSEDING
Fed. Spec. DDD-W-80C
June 22, 1971

FEDERAL SPECIFICATION

WASHCLOTH, TERRY, COTTON

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for two types of cotton washcloths.

1.2 Classification.

1.2.1 Types. The cotton washcloth shall be of the following types as specified (see 6.2):

Type I - 16 ounce per dozen (min)

Type II - 11 ounce per dozen (min)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

V-T-276	-	Thread, Cotton.
UU-P-268	-	Paper, Kraft, Wrapping.
DDD-L-20	-	Label; For Clothing, Equipage, and Tentage, (General Use).
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-T-45	-	Tape, Gummed Paper, Reinforced and Plain, For Sealing and Securing.

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Federal Standards:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).
Fed. Test Method Std. No. 191 - Textile Test Methods.
Fed. Std. No. 751 - Stitches, Seams, and Stitchings.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Fransicso, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads for 40" x 48" Pallets.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

16CFR303 - Rules and Regulations Under the Textile Fiber Products Identification Act.

(Copies may be obtained without charge from the Federal Trade Commission, Washington, DC 20580.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

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National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Standard sample. When a standard sample is available, the finished washcloth shall match the standard sample for shade, and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 Material.

3.2.1 Yarns. The yarns shall be single or two ply yarns as specified in table I. The filling and pile warp yarns shall be made of cotton. The ground warp yarns shall be cotton or polyester/cotton which may contain up to 70 percent polyester with the remaining percentage cotton.

3.3 Physical requirements. The finished washcloths shall conform to the requirements specified in table I when tested as specified in 4.4.

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TABLE I. Physical requirements

Type	Weight per dozen (minimum)	Yarns per inch 1/ Warp (min.)	Yarns per inch Filling (min.)	Ply warp Ground	Ply warp Pile	Filling Ply	Finished dimensions Each	Finished dimensions Tolerance	Break strength (minimum) Warp	Break strength (minimum) Filling
	Ounce						Inch	Inch	Lbs.	Lbs.
I	16	68 2/	35 2/	2 or 1	2 or 1	1	12 by 12	± 1/2	40	35
II	11	59	29	2 or 1	2 or 1	1	12 by-12	± 1/2	25	25

1/ The washcloths shall reach equilibrium under Standard Conditions as defined in FED-STD-191 prior to determination of weight.

2/ Either of these values may fall below this minimum by not more than three yarns, provided the combined warp and filling yarn total does not fall below 100 yarns per inch.

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3.3.1 Weave.

3.3.1.1 Type I. The type I washcloth shall consist of a ground warp, pile warp, and filling yarns woven as a single-loop terry weave except that there shall be a plain woven border 1/4 to 1/2 inch wide either on both sides and both ends or on both ends only. When the border is on both ends only, it shall be located not more than 1 inch from the ends.

3.3.1.2 Type II. The type II washcloth shall consist of a ground warp, pile warp, and filling yarns woven as single-loop terry weave. The border shall be on both sides and both ends, or on both ends only. The border measurement, from the edge towards the body, shall be in this order: 3/8 to 5/8 inch plain weave, 1/8 to 3/8 inch terry, 1/16 to 3/16 inch plain weave, 1/8 to 3/8 inch terry; 1/16 to 3/16 inch plain weave, terry body.

3.3.2 Thread. The thread for stitching the type I washcloth shall be cotton, ticket No. 30, 3 ply, type IA2 of V-T-276. The thread for stitching the type II washcloth shall be cotton, ticket No. 50, 3 ply, type IA2 of V-T-276.

3.4 Construction.

3.4.1 Edge finishing. All four edges shall be overedged stitched with 10 to 14 stitches per inch of stitch type 504 or 505 and stitching type EFd-1 as specified in FED-STD-751. All ends of stitching, thread breaks, skipped stitches and runoffs shall be overstitched 3/4 to 1 inch. There shall be no loose or untrimmed threads on the finished washcloths.

3.5 Color. The washcloths shall be desized and bleached white to match the standard sample when available. The bleached washcloths shall show no more yellowing or discoloration than that shown by the standard sample after 20 hours exposure to light (see 4.4). When no standard sample is available, the discoloration shall be no more than the rating of "Fair" when tested as specified in 4.4.

3.5.1 Matching. The bleached washcloths shall match the standard sample under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin.

3.6 Nonfibrous material content. The starch and protein content including chloroform-soluble and water-soluble material of the finished cloth shall not exceed 2.0 percent when tested as specified in 4.4.

3.7 Marking for identification. When marking for identification is specified (see 6.2), each washcloth shall have an identification label conforming to the requirements of type I, class 4 of DDD-L-20, except that all references to size shall be deleted. The right side of the label shall

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be caught in the overedge stitching and the bottom edge of the label shall be 1/2 to 3/4 inch from the lower corner of the washcloth. The end item is intended to be laundered.

3.7.1 Fiber identification. Each washcloth shall be labeled, for fiber content in accordance with the Textile Fiber Products Identification Act.

3.8 Workmanship. The finished washcloths shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the levels set by the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.2.2 Inspection of the end item.

4.2.2.1 Examination. The defects found during examination shall be classified in accordance with 4.2.2.1.1, 4.2.2.1.2, and 4.2.2.1.3.

4.2.2.1.1 Visual examination. The inspection level shall be level I of MIL-STD-105 and the acceptable quality level (AQL) shall be 4.0 percent defective.

Defects

Cut, hole, tear, or smash.
Three or more contiguous broken or missing yarns (clearly visible).
Terry loops improperly formed over an area of 1 square inch or bald areas of 1/2 square inch or more.
Spot, stain, streak, clearly visible.
Any trash, motes, leaf, or seed coat fragments.

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Defects (cont'd

Objectionable odor.
Color or shade not as specified.
Wrong stitch type or stitches per inch not as specified.
Any overedge stitching missing, three or more consecutively skipped stitches, stitching runoffs, loose stitch tension resulting in a loosely secured edge.
Identification label missing, incorrect, illegible, insecurely attached or misplaced.
Border not as specified.
Any thread breaks, runoffs and ends of stitching not overstitched 3/4 to 1 inch.

4.2.2.1.2 Dimensional examination. The sample size shall be in accordance with inspection level S-3. The sample unit for the examination shall be one washcloth. The AQL shall be 4.0 percent defective.

Overall length and width defects:
Less than 11-1/2 inches
More than 12-1/2 inches

4.2.2.1.3 Examination for weight. The sample unit shall be one dozen washcloths. Any unit that does not meet the minimum weight requirements specified in table I shall be considered a defect. The inspection level shall be S-1 and the AQL shall be 4.0 percent defective.

4.3 Examination for preparation for delivery requirements. An examination shall be made to determine that packaging, packing and marking comply with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be closed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 defects per hundred units.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence or method of application.
Materials	Any component missing, damaged or not as specified.

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Examine	Defect
Workmanship	Wrap not secured as specified, incomplete closure of container flaps, loose strapping, improper taping, inadequate stapling. Bulged or distorted containers. Open and noncontinuous heat sealed seams and closure of polyethylene bag. Omission of vent hole in polyethylene bag. Incorrectly fabricated bag.
Content	Number of bundles or bags per container is more or less than specified. Number of washcloths per bundle or bag is more or less than specified. 1/

1/ For this defect, three bundles or bags shall be examined from each shipping container in the sample.

4.3.1 Examination for palletization. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully prepared for delivery. The lot size shall be the number of palletized unit loads in the end item inspection lot. The inspection level shall be S-1 and the AQL shall be 6.5 defects per 100 units.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified minimum requirement.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence or method of application.

4.4 Testing of the end item. The methods of testing specified in FED-STD-191 wherever applicable and as listed in table III shall be followed. The physical and chemical values specified in section 3 apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test method. The sample unit shall be five washcloths. The sample size shall be in accordance with table II. The lot size shall be expressed in units of one washcloth each. The lot shall be unacceptable if one or more sample units fail to meet a test requirement.

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TABLE II. Sampling for tests

Lot size	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and up	5

TABLE III. Test methods

Characteristic	Requirement paragraph	Test method
Yarns per inch	3.3	5050
Breaking strength	3.3	5100
Weave	3.3.1	Visual 1/
Discoloration	3.5	5660 1/ 2/
Nonfibrous material	3.6	2611

1/ One determination shall be made on each sample unit and the results reported as "pass" or "fail".

2/ Method 5660 shall be used, except that exposure of the specimen and standard sample shall be for 20 standard fading hours. Any specimen showing discoloration less than or equal to that of the standard sample shall be rated "pass". Any specimen showing discoloration greater than that of the standard sample shall be rated "fail". When no standard sample is available, the discoloration of the specimen shall be considered satisfactory if it is not worse than the rating of "fair" indicated in Method 5660.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. Twelve washcloths of one type only shall be evenly stacked, flat or center-folded, and then completely wrapped in 30-pound minimum basis weight kraft paper conforming to type I, grade B of UU-P-268. Each paper wrapped bundle shall be cross-tied with cotton tape or twine, or secured with gummed paper tape conforming to type III, grade B of PPP-T-45. Alternatively, the stack of twelve washcloths shall be packaged in a close-fitting clear polyethylene film bag of 0.00125 inch thickness (± 25 percent tolerance). The polyethylene bag

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shall be formed with heat sealed seams that are straight, continuous, and parallel to each other and the formed edges of the bag. The bag closure shall be effected by heat sealing with the heat seal made as close as possible to the open end or by means of a mechanical tie (paper or vinyl plastic covered soft steel wire, aluminum band, etc.). A 1/4 inch maximum diameter hole shall be made at one corner of each polyethylene bag to allow the excess air to escape. The bag may be fabricated from polyethylene film tubing or sheeting.

5.1.2 Level C. Washcloths shall be packaged to afford adequate protection against physical damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Three hundred washcloths of one type only, packaged as specified in 5.1, shall be compactly packed within a snug-fitting fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636. The inside of each container shall be fitted with a box liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V and reinforced as specified in the appendix of PPP-B-636.

5.2.2 Level B. Three hundred washcloths of one type only, packaged as specified in 5.1, shall be compactly packed within a snug-fitting fiberboard shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. The inside of each container shall be fitted with a taped box liner conforming to class domestic, variety SW, grade 275 of PPP-B-636. Each shipping container shall be closed in accordance with method II as specified in the appendix of the container specification.

5.2.2.1 When specified (see 6.2), the fiberboard shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of the container specification.

5.2.3 Level C. Washcloths, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transporation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification or National Motor Freight Classification, as applicable.

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5.3 Palletization. When specified (see 6.2), washcloths of one type only shall be palletized in accordance with load type I of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

5.4 Marking. Marking shall be in accordance with 5.4.1 or 5.4.2 as specified (see 6.2).

5.4.1 Civil agencies. In addition to any special marking required by the contract or order, interior packages, shipping containers and palletized unit loads shall be marked in accordance with FED-STD-123.

5.4.2 Military requirements. In addition to any special marking required by the contract or order, interior packages, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129.

5.4.3 Polyethylene bagged packages. Polyethylene bagged packages shall have the required identification information legibly printed or stamped in black directly on the bag across the center face or on a white paper label inserted within the bag so as to permit ready identification.

6. NOTES

6.1 Intended use. The washcloths are intended for use by the Federal agencies and the Department of Defense.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

(a) Title, number, and date of this specification.
(b) Type of washcloth required (see 1.2).
(c) When marking identification is required (see 3.7).
(d) Selection of the applicable levels of packaging and packing (see 5.1 and 5.2).
(e) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
(f) When palletization is required (see 5.3).
(g) Marking required (see 5.4).

6.3 Standard sample. For access to standard sample, address the procuring office issuing the invitation for bids (see 3.1).

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Army - GL
Navy - SA
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Review activities:

Army - MD
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Preparing activity:

Army - GL

Civil Agency Coordinating Activities:

GSA - FSS

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